WORKED SOLUTION

STEP II 2024, Question 12

Probability and Statistics • Discrete Order Statistics

STEP II 2024, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

A winner in the first game is a player holding a *unique* smallest number, and there can be at most one such player. So "there is a winner" splits into disjoint cases according to who wins and with which number, and each case is a simple product of independent probabilities. The given formulae for $\sum i^2$ and $\sum i^3$ are the hint that the resulting sums are polynomial sums. The second game works the same way, now with an ordered pair of players (one lowest, one highest) and two numbers to sum over.

Solution

Throughout, the k numbers are independent and each is uniform on $\{1, 2, \ldots, n\}$, so any particular player receives any particular number with probability $\frac{1}{n}$, and receives a number in a given set of s values with probability $\frac{s}{n}$.

Part (i)

Fix a with $1 \leq a \leq n-1$. Ada must receive a (probability $\frac{1}{n}$), and each of the other $k-1$ players must receive one of the $n-a$ numbers $a+1, \ldots, n$ (probability $\frac{n-a}{n}$ each). By independence,

$$\mathrm{P}(\text{Ada gets } a, \text{ all others larger}) = \frac{1}{n}\left(\frac{n-a}{n}\right)^{k-1} = \frac{(n-a)^{k-1}}{n^k}.$$

(For $a = n$ this would be zero, which is why a stops at $n-1$.)

Let W be the event that there is a winner. A winner holds a number strictly smaller than everyone else's, so there cannot be two winners: each would have to be smaller than the other. Hence W is the disjoint union, over the k players and over $a = 1, \ldots, n-1$, of the events "this player gets a and all the others get larger numbers". By symmetry each

player contributes the same amount as Ada, so

$$\mathrm{P}(W) = k\sum_{a=1}^{n-1}\frac{(n-a)^{k-1}}{n^k} = \frac{k}{n^k}\sum_{j=1}^{n-1} j^{k-1},$$

where we substituted $j = n-a$, which also runs from 1 to $n-1$.

With $k=4$, using the given formula for $\sum i^3$ with $n-1$ in place of n,

$$\mathrm{P}(W) = \frac{4}{n^4}\cdot\frac{1}{4}(n-1)^2n^2 = \frac{(n-1)^2}{n^2},$$

as required.

Part (ii)

Fix d and a with $1 \le d \le n-2$ and $1 \le a \le n-d-1$; these conditions guarantee that $a+d+1 \le n$ and that there is at least one number strictly between a and $a+d+1$. Ada must receive a and Bob must receive $a+d+1$ (probability $\frac{1}{n}$ each), and each of the other $k-2$ players must receive one of the d numbers $a+1,\ldots,a+d$. By independence,

$$\mathrm{P}(\text{Ada gets } a,\ \text{Bob gets } a+d+1,\ \text{others strictly between}) = \frac{1}{n^2}\left(\frac{d}{n}\right)^{k-2} = \frac{d^{k-2}}{n^k}.$$

Ask yourself Why does every way of getting two winners appear exactly once in a sum over players and over a and d?

Two winners means one player has a unique smallest number and a different player has a unique largest number. As in part (i), the low winner is unique and so is the high winner, so the outcome determines an *ordered* pair of distinct players (low winner, high winner) together with the low number a and the gap $d \ge 1$ to the high number: the gap must be at least 1 because the $k-2 \ge 2$ other players' numbers lie strictly between. Different choices of these give disjoint events. There are $k(k-1)$ ordered pairs of players, and each contributes the same as (Ada, Bob), so

$$\mathrm{P}(\text{two winners}) = k(k-1)\sum_{d=1}^{n-2}\sum_{a=1}^{n-d-1}\frac{d^{k-2}}{n^k} = \frac{k(k-1)}{n^k}\sum_{d=1}^{n-2}(n-1-d)\,d^{k-2},$$

since the inner sum has $n-d-1$ equal terms.

Now take $k=4$ and write $m = n-1$, so d runs from 1 to $m-1$. Using both given formulae with $m-1$ in place of n,

$$\begin{aligned}\sum_{d=1}^{m-1}(m-d)\,d^2 &= m\cdot\tfrac{1}{6}(m-1)m(2m-1) - \tfrac{1}{4}(m-1)^2m^2\\ &= \tfrac{1}{12}m^2(m-1)[2(2m-1)-3(m-1)]\\ &= \tfrac{1}{12}m^2(m-1)(m+1).\end{aligned}$$

Therefore, since $m-1=n-2$ and $m+1=n$,

$$\mathrm{P}(\text{two winners}) = \frac{12}{n^4}\cdot\frac{1}{12}(n-1)^2(n-2)n = \frac{(n-2)(n-1)^2}{n^3},$$

which is the required result.

Comparing one and two winners when $k=4$

Let L be the event that some player has a unique smallest number and H the event that some player has a unique largest number. Part (i) gives $\mathrm{P}(L)=\frac{(n-1)^2}{n^2}$. Replacing every number x by $n+1-x$ turns a uniform number into a uniform number and swaps "smallest" with "largest", so $\mathrm{P}(H)=\mathrm{P}(L)$ too.

Exactly two winners is the event $L\cap H$, so we write $T=\mathrm{P}(L\cap H)=\frac{(n-2)(n-1)^2}{n^3}$. Exactly one winner means L or H happens but not both, so

$$\mathrm{P}(\text{exactly one}) = \mathrm{P}(L)+\mathrm{P}(H)-2\,\mathrm{P}(L\cap H) = \frac{2(n-1)^2}{n^2}-2T.$$

Two winners are more likely than one exactly when

$$\begin{aligned} T > \frac{2(n-1)^2}{n^2}-2T &\iff 3\cdot\frac{(n-2)(n-1)^2}{n^3} > \frac{2(n-1)^2}{n^2} \\ &\iff 3(n-2)>2n \iff n>6, \end{aligned}$$

where we divided by the positive quantity $\frac{(n-1)^2}{n^3}$ (positive because $n\geq 3$). At $n=6$ the two probabilities are equal, so the minimum value is

$$n=7.$$

Checks

- For $n=3$ and $k=4$ there are 81 outcomes. A winner with a 1 needs the other three numbers in $\{2,3\}$: $4\cdot 2^3=32$ outcomes. A winner with a 2 needs three 3s: 4 outcomes. So $\mathrm{P}(W)=\frac{36}{81}=\frac{4}{9}=\frac{(3-1)^2}{3^2}$.
- At $n=6$ both "exactly one" and "exactly two" have probability $\frac{25}{54}\approx 0.463$; at $n=7$ they are $\frac{144}{343}\approx 0.420$ and $\frac{180}{343}\approx 0.525$. These, and the formulae in both parts for $n\leq 8$ and $k=2,3,4$, were confirmed by a computer enumeration of every outcome.
- As $n\to\infty$ ties become rare, so with four players there should almost always be both a unique lowest and a unique highest: indeed $\frac{(n-2)(n-1)^2}{n^3}\to 1$.

Common slips

- Forgetting the factor k in part (i), or the factor $k(k-1)$ in part (ii). In part (ii) the pair of winners is ordered, since one is low and one is high.

- Computing "exactly one winner" as $\mathrm{P}(L) + \mathrm{P}(H) - \mathrm{P}(L \cap H)$, which is "at least one"; the overlap must be subtracted twice.
- Answering $n = 6$: there the two probabilities are equal, not strictly ordered.

Beyond STEP

Splitting an event into disjoint pieces labelled by "who" and "where" is the discrete version of finding the distribution of order statistics, the smallest and largest of a sample. For continuous samples the same argument gives the joint density of the minimum and maximum, and the ties that make this question interesting disappear. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet these ideas.

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